

Ningzhe Shi

Ph.D. in Computer Science and Technology

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Research Profile

Ph.D. researcher in 6G intelligent networks and AI service orchestration, with a focus on communication, sensing, computing, memory, and knowledge-resource optimization through reinforcement learning and mathematical optimization.

Education

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| <p>Ph.D. Institute of Computing Technology, Chinese Academy of Sciences; University of Chinese Academy of Sciences, Computer Science and Technology (A+ national discipline)</p> <ul style="list-style-type: none"> Ph.D. Advisor: Prof. Yiqing Zhou | <p>Beijing, China
Sept 2021 – present</p> |
| <p>B.Eng. Xidian University, Communication Engineering (A+ national discipline)</p> <ul style="list-style-type: none"> Undergraduate Advisor: Prof. Junyu Liu Ranked 1st in class National Scholarship recipient | <p>Xi'an, China
Sept 2017 – July 2021</p> |

Internship Experience

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| <p>China Mobile, Digital Intelligence Department, AI Agent Algorithm Intern</p> <ul style="list-style-type: none"> Investigated long-horizon memory mechanisms for AI agents in multi-turn interactions, focusing on context forgetting, historical redundancy, and response consistency. Surveyed retrieval-augmented memory, hierarchical storage, self-evolving memory updates, memory extraction, compression, forgetting, and dynamic updating. Reproduced and analyzed the Mem-alpha memory architecture and formulated memory selection, compression, and updating as a policy-optimization problem. Studied reinforcement-learning-based memory maintenance and memory-aware LLM task scheduling under end-edge-cloud collaboration. | <p>China
Jan 2026 – Mar 2026</p> |
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Research Experience

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| <p>Institute of Computing Technology, Chinese Academy of Sciences, Main Contributor, Knowledge-Aware Joint Scheduling and Retrieval Activation for RAG Services</p> <ul style="list-style-type: none"> Modeled task semantics, server-side knowledge coverage, retrieval activation, computing cost, transmission delay, and energy consumption for cloud-edge RAG/GAI services. Designed a knowledge-aware joint scheduling and retrieval activation mechanism under heterogeneous knowledge bases and dynamic resource states. Showed that adaptive retrieval activation can reduce average service delay by 67% and 62% compared with retrieval-always-on and retrieval-always-off baselines, respectively. | <p>Oct 2025 – June 2026</p> |
| <p>Institute of Computing Technology, Chinese Academy of Sciences, Main Contributor, Memory-Aware Distributed Routing for Space AI Computing Networks</p> <ul style="list-style-type: none"> Modeled the coupling among task routing, node memory state, onboard computing load, storage availability, and link conditions. Designed a distributed routing method for high-dimensional dynamic network states and introduced diffusion-assisted policy generation. Validated the method on a ground-based space computing network simulator, reducing end-to-end routing delay by more than 15%. | <p>Oct 2025 – Mar 2026</p> |

Institute of Computing Technology, Chinese Academy of Sciences , Main Contributor, Distributed Quantum Reinforcement Learning for AIGC Cloud-Edge Scheduling - Modeled collaborative cloud-edge processing for multi-user AIGC services, coupling task offloading, computing allocation, and service delay. - Designed a distributed hybrid quantum-classical reinforcement learning scheduler using parameterized quantum circuits and classical neural networks. - Reduced model parameters by more than 95% and average service delay by more than 30% compared with baseline methods in simulation.	June 2025 – Dec 2025
Institute of Computing Technology, Chinese Academy of Sciences , Main Contributor and First Author, ISAC-Driven AIGC Service Orchestration - Designed a content-accuracy-and-quality-aware service metric and modeled sensing, computing, communication, and AIGC service experience as a coupled system. - Formulated the joint resource-allocation problem and designed an LP-guided hierarchical reinforcement learning framework. - Improved average user service experience by more than 50% in evaluated settings; related work was published in IEEE Transactions on Mobile Computing.	Jan 2024 – June 2025
Institute of Computing Technology, Chinese Academy of Sciences , Main Contributor and First Author, User Selection and Resource Allocation for High-Concurrency AI Services - Modeled user access, bandwidth, subchannels, transmit power, edge computing resources, and service satisfaction in multi-cell MEC networks. - Designed a joint optimization algorithm for user selection, channel assignment, power control, and edge computing resource allocation. - Improved average user satisfaction by more than 50%; related work was published in IEEE Transactions on Mobile Computing.	Dec 2022 – Dec 2024
Institute of Computing Technology, Chinese Academy of Sciences , Main Contributor and First Author, Adaptive Fair Resource Scheduling Under Imperfect CSI - Built a channel-uncertainty-aware resource allocation model for multi-cell multi-user downlink communications under imperfect CSI. - Designed a reward-adaptive multi-agent DRL scheduler balancing long-term fairness and network service rate. - Improved long-term fairness by 13% and service rate by more than 10%; related work was published in IEEE Transactions on Cognitive Communications and Networking.	May 2022 – May 2024
Institute of Computing Technology, Chinese Academy of Sciences , Technical Lead, Multi-Channel Access for In-Band Full-Duplex Ad Hoc Networks - Designed multi-channel interference coordination and access-control mechanisms for full-duplex ad hoc networks with residual self-interference and inter-node interference. - Built a distributed WebSocket-based system-level simulator integrating node workers, channel computation, mobility modeling, and visualization. - Delivered a system prototype and related invention patent/software copyright outputs.	Oct 2022 – June 2023
Institute of Computing Technology, Chinese Academy of Sciences , Task Leader, Technical Lead, and First Author, UAV-Assisted Secure Transmission Under Deception and Jamming - Built a UAV-assisted mmWave MIMO secure communication model covering legitimate links, deception links, jamming sources, and UAV-assisted nodes. - Designed joint optimization for beamforming, UAV three-dimensional placement, and power allocation under complex electromagnetic confrontation. - Improved secure transmission rate to approximately two times that of the no-deception baseline; related work was published in China Communications and one invention patent was granted.	Sept 2021 – June 2023

Publications

- Service Satisfaction Based User Selection and Resource Allocation for NOMA-Based Multi-Cell MEC Networks** 2026
Ningzhe Shi, Yiqing Zhou, Ling Liu, Yihao Wu, Hanxiao Yu, Jinglin Shi
[10.1109/TMC.2026.3673357](#) (IEEE Transactions on Mobile Computing)
- Content Accuracy and Quality Aware Resource Allocation Based on LP-Guided DRL for ISAC-Driven AIGC Networks** 2026
Ningzhe Shi, Yiqing Zhou, Ling Liu, Jinglin Shi, Yihao Wu, Haiwei Shi, Hanxiao Yu
[10.1109/TMC.2026.3652144](#) (IEEE Transactions on Mobile Computing)
- Long-Term Fairness From Real-Time Decisions: Reward Adaptive Multi-Agent DRL for Wireless Communication With Imperfect CSI** 2026
Ningzhe Shi, Ling Liu, Yiqing Zhou, Hanxiao Yu, Yihao Wu, Jingya Yang, Jinglin Shi
[10.1109/TCCN.2026.3692166](#) (IEEE Transactions on Cognitive Communications and Networking)
- Deception-Based Secure Communication for UAV-Assisted mmWave MIMO Systems** 2026
Ningzhe Shi, Yiqing Zhou, Yu Zhang, Zhijun Han, Ling Liu, Jinglin Shi
[10.23919/JCC.fa.2025-0337.202604](#) (China Communications)
- Cross-Enabling of Operation, Information and Communication: Building the Foundation for Intelligent New-Generation Productivity** 2025
Ningzhe Shi, Yihao Wu, Yaxing Xu, Qing Cai, Yiqing Zhou
[10.11991/cccf.202508013](#) (Computing Magazine of the CCF)
- Deep Reinforcement Learning Based Subchannel Selection and Power Allocation in Wireless Networks with Imperfect CSI** 2023
Ningzhe Shi, Yu Zhang, Yiqing Zhou
(2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring))
- Performance Analysis and Energy-Efficient Optimization for Multiple Active RIS-Assisted MIMO Communication Systems** 2026
Hanxiao Yu, *Ningzhe Shi*, Yiqing Zhou, Zhong Zheng, Jinglin Shi, Zesong Fei
(IEEE Transactions on Vehicular Technology)
- OFDM-Based NOMA-Enhanced Coexistence of SemCom and BitCom System** 2025
Hanxiao Yu, *Ningzhe Shi*, Jinglin Shi, Jing Guo, Siqiang Wang, Zesong Fei
(IEEE Wireless Communications Letters)
- Cost-Aware Deep Reinforcement Learning for eMBB-URLLC Multiplexing Resource Allocation** 2025
Z. Lin, *Ningzhe Shi*, Yiqing Zhou, Jinglin Shi
(2025 IEEE Wireless Communications and Networking Conference (WCNC))

Granted Invention Patents

- Construction Method and Resource Management Method for a Wireless Network Resource Allocation System** Apr 2026
Granted invention patent: CN202310354794.0; authorized 2026-04-07
- Anti-Jamming Method and System Based on Relay Position Selection and Power Allocation** May 2025
Granted invention patent: CN202310078808.0; authorized 2025-05-27

Anti-Tracking Jamming-Resistant Signal Transmission Method, Signal Reception Method, and System	Sept 2024
Granted invention patent: CN202211134854.X; authorized 2024-09-24	

Invention Patent Applications

Transmission Device and Semantic Transmission Method for High-Mobility Communications	Mar 2026
Invention patent application: CN202610406457.5; filed 2026-03-31	
Joint Channel Estimation and Signal Detection Method, Signal Receiver, and Storage Medium	Oct 2025
Invention patent application: CN202511536493.5; filed 2025-10-27	
Resource Allocation Method for AI-Generated Content Services Based on Integrated Sensing and Communication	July 2025
Invention patent application: CN202511053332.0; filed 2025-07-30	
Environment- and Resource-State-Aware Resource Allocation Method and System	July 2025
Invention patent application: CN202510955830.8; filed 2025-07-11	
Mobile Ad Hoc Network Data Routing Method and Industrial Ad Hoc Network System	June 2025
Invention patent application: CN202510873622.3; filed 2025-06-27	
Channel Access Method and Communication System for Ad Hoc Networks	Mar 2024
Invention patent application: CN202410348337.5; filed 2024-03-26	
Multi-User MIMO Signal Detection Model	Jan 2024
Invention patent application: CN202410018551.4; filed 2024-01-05	
Real-Time Simulation System and Method for Full-Duplex Ad Hoc Networks	Dec 2023
Invention patent application: CN202311719290.0; filed 2023-12-14	
Construction Method and Resource Management Method for a Wireless Network Resource Allocation System	Apr 2023
Invention patent application (PCT): PCT/CN2023/090283; filed 2023-04-24	

Software Copyright

Large-Scale Full-Duplex Network Simulation System	Jan 2024
Registered software copyright: 2024SR0082117; registered 2024-01-11	

Honors and Awards

Outstanding Graduate Student, Xinghang Program	June 2026
Journal of Electronics & Information Technology; national selection of 22 graduate students	
Outstanding Poster Award	June 2026
Frontiers in Electronics and Information Conference	
UCAS Outstanding Student Model	June 2026
University-wide top 1%	
ICT Outstanding Student	Jan 2026
Institute of Computing Technology / State Key Laboratory of Processors	
E Fund Financial Technology Doctoral Scholarship	Jan 2025
12 recipients institute-wide	

UCAS Outstanding Student Cadre	May 2025
Outstanding Volunteer, Chinese Academy of Sciences Public Science Day	2023-2025
UCAS Outstanding Student	June 2023
UCAS First-Class Scholarship Awarded for five consecutive years	2021-2025
Outstanding Graduate, Xidian University	June 2021
First-Class Graduation Scholarship, Xidian University	June 2021
Honorable Mention, Mathematical Contest in Modeling	2020
National Scholarship	2019
National Second Prize, China Undergraduate Mathematical Contest in Modeling	2019
National Encouragement Scholarship	2018
Third Prize, CETC-10 Communication Cup	2019

Academic Service

Program Committee: Technical Program Committee Member, IEEE International Conference on Communications (ICC), 2026

Academic Correspondence: Academic Correspondent, Digital Communications and Networks

Peer Review: IEEE Transactions on Mobile Computing (TMC); IEEE Transactions on Communications (TCOM); IEEE Internet of Things Journal; IEEE Transactions on Vehicular Technology (TVT); Science China Information Sciences; China Communications; Journal of Information and Intelligence; IEEE International Conference on Communications (ICC); IEEE Vehicular Technology Conference (VTC); Journal on Communications; Journal of Electronics & Information Technology

Outreach and Public Engagement

Science Outreach

- Science outreach volunteer, Chinese Academy of Sciences Public Science Day
- Committee Member, UCAS Science Popularization Association
- Participated in public science communication, research demonstrations, and science outreach activities
- Outstanding Science Popularization Volunteer, Bureau of Academic Divisions, Chinese Academy of Sciences

Leadership and Student Service

Graduate Stage

- Vice President, Haidian Branch of the Volunteer Federation
- Committee Member, UCAS Science Popularization Association
- Class Mental Health Liaison

Undergraduate Stage

- Youth League Branch Secretary
- Team Leader, Horizon Volunteer Team, School of Telecommunications Engineering, Xidian University

Skills

AI-Native Network Research: 6G intelligent networks; cloud-edge collaborative AI services; RAG service orchestration; AI Agent memory systems; space AI computing networks; ISAC-driven AIGC services

Optimization and Learning: Reinforcement learning; multi-agent DRL; LP-guided hierarchical scheduling; diffusion-assisted policy generation; quantum-classical reinforcement learning; convex optimization; robust and fairness-aware resource allocation

Wireless and Computing Systems: MEC/NOMA networks; full-duplex ad hoc networks; UAV-assisted secure communications; integrated sensing, communication, computing, and knowledge-resource optimization

Implementation and Simulation: Python; PyTorch; MATLAB; C++; Shell; custom RL environments; system-level wireless simulation; distributed WebSocket-based simulation; experiment automation and visualization

Research Delivery: Problem formulation; algorithm design; simulation validation; ablation studies; technical reporting; journal paper writing; patent and software-copyright preparation; cross-team technical coordination